AtChem2 v1.3

User Manual

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Chapter 1

Introduction

AtChem2 is an open-source modelling tool for atmospheric chemistry. It is designed to build and run zero-dimensional box-models using the Master Chemical Mechanism (MCM). The **MCM** is a near-explicit chemical mechanism which describes the gas-phase oxidation of primary emitted volatile organic compounds (VOC) to carbon dioxide (CO_2) and water (H_2O). The MCM protocol is detailed in Jenkin et al. [1997] and subsequent updates [Saunders et al., 2003, Jenkin et al., 2003, Bloss et al., 2005, Jenkin et al., 2012, 2015]. AtChem2 is designed for use with the MCM, but it can use any other chemical mechanism, as long as it is provided in the correct format (Sect. 3.1).

AtChem2 is a development of **AtChem-online**, a modelling web tool created at the University of Leeds to facilitate the use of the MCM in the simulation of laboratory and environmental chamber experiments within the EUROCHAMP community [Martin, 2009]. AtChem-online runs as a web service, provided by the University of York; to use AtChem-online, the user needs only a text editor, a file compression tool, a web browser, and an internet connection. A tutorial – with examples and exercises – is available on the MCM website.

AtChem-online is easy to use even for inexperienced users but has a number of limitations, mostly related to its nature as a web application. **AtChem2** has been developed from the AtChem-online codebase with the aim of providing an *offline* modelling tool capable of running long simulations of computationally intensive models, as well as batch simulations (e.g., for sensitivity studies). In addition, AtChem2 implements a continuous integration workflow, along with version control (using git) and a comprehensive Testsuite, which makes it robust, reliable and traceable.

The AtChem2 codebase is structured into four components (or layers), as outlined in Fig. 1.1, and is written in Fortran 90/95. Installation, compilation and execution of AtChem2 are semi-automated via a number of shell and Python scripts that require minimal input from the user. Model con-

figuration and inputs consist of simple text files that can be easily modified with any text editor.

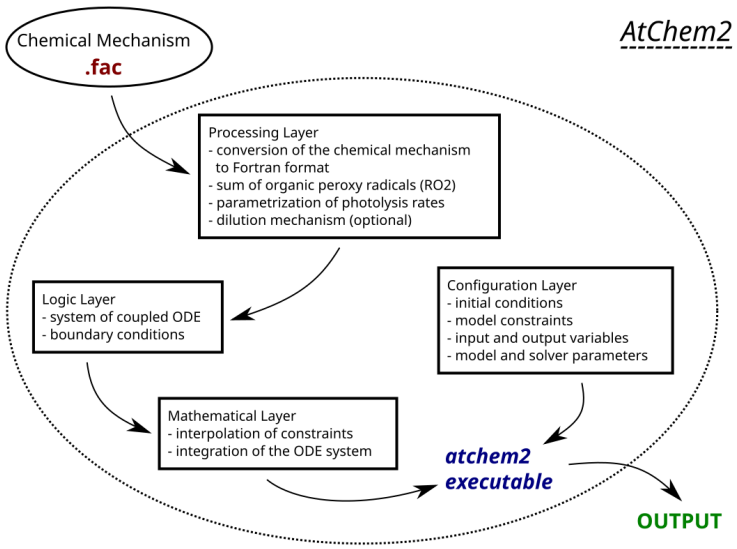


Figure 1.1: Architecture of AtChem2.

This document (`AtChem2-Manual.pdf`) is the AtChem2 user manual and contains all the information required to install, configure, and use the current version of the model. The AtChem2 wiki contains summaries of these instructions for quick user reference. Additional information, such as example submission scripts for high-performance computing (HPC) systems, and guidance for resolving some known issues can also be found on the wiki.

AtChem-tools is a set of tools designed for AtChem2, available at: https://github.com/AtChem/AtChem-tools. The tools are written in Python and include scripts to help with the model setup and configuration, to run the model, and to analyze the model outputs. For instructions, see the AtChem-tools documentation.

1.1 Conventions

The following conventions are adopted in the AtChem2 user manual and documentation:

`Main Directory` is the base directory of AtChem2.
For example: `$HOME/AtChem2/`.

`Dependencies Directory` is the directory where the AtChem2 dependencies (Sect. 2.3) are installed.
For example: `$HOME/AtChem-lib/`.

`Model Directory` is the directory containing the configuration, initial conditions, chemical mechanism and constraints, and model output. For example: `$HOME/AtChem2/model/`.

The user can change the locations and names of these directories according to their setup, workflow, personal preferences, etc. As long as the correct paths to the libraries and directories needed by AtChem2 are specified, the model will compile and run.

Important Note: AtChem2 runs in **Coordinated Universal Time (UTC)**, with no daylight saving. This is to simplify the calculation of the solar angles and, therefore, of the photolysis rates (Sect. 3.5). The user must ensure that the input data and the model constraints are provided in UTC (i.e. GMT timezone) and convert the output to Local Time, if needed.

1.2 License and Citation

AtChem2 is open-source and free to use, under the terms of the **MIT license**. A copy of the license can be found in the `LICENSE` file. If used in a publication, please cite the GMD paper [Sommariva et al., 2020]:

R. Sommariva, S. Cox, C. Martin, K. Borońska, J. Young, P. K. Jimack, M. J. Pilling, V. N. Matthaios, B. S. Nelson, M. J. Newland, M. Panagi, W. J. Bloss, P. S. Monks, and A. R. Rickard. **AtChem (version 1), an open-source box model for the Master Chemical Mechanism**. *Geoscientific Model Development*, 13, 1, 169–183, 2020. DOI: 10.5194/gmd-13-169-2020.

Chapter 2

Model Installation

2.1 Requirements

AtChem2 can be installed and used on Linux/Unix or macOS operating systems. A working knowledge of the **unix shell** and its basic commands is *required* to install and use the model. AtChem2 requires the following software tools:

- ◇ a terminal with bash or another sh-compatible shell (e.g., zsh, which is the default shell in macOS), and common unix utilities: wget, tar, zip, make, etc.
- ◇ a text editor, such as Emacs, vim, Gedit, Kate, TextEdit, etc.
- ◇ a Fortran compiler – the model compiles with GNU `gfortran` (version 4.8 or later) and with Intel `ifort` (version 17.0 or later).
  The default Fortran compiler is `gfortran`.
- ◇ a Python installation[1].

Some or all of these tools may already be installed on the operating system: use the `which` command to find out (e.g., `which python`, `which make`). Otherwise, check the local documentation or ask the system administrator. In addition, AtChem2 has the following dependencies:

- ◇ CVODE – requires gcc, cmake
- ◇ openlibm
- ◇ numdiff (optional)
- ◇ FRUIT (optional) – requires Ruby v3.0 or earlier

[1] Python v3 is recommended, but all the scripts used in AtChem2 also work with Python v2. Support for Python v2 will be removed in future versions.

For detailed instructions on the installation of the dependencies, go to Sect. 2.3. AtChem2 can also be run as a Docker container: this is still an *experimental feature* and may require some user intervention to make it work properly. For more information, go to Sect. 2.5.

2.2 Download

The primary repository for the AtChem2 code and documentation is: https://github.com/AtChem/AtChem2.

Two versions of the model are available:

1. the *stable version*, provided as a compressed archive file (`.tar.gz` or `.zip`) on the Releases page of the repository.
2. the *development version*, corresponding to the `master branch` of the repository.

Although the two versions are usually equivalent, it is strongly recommended to use the *stable version* for normal work, because it is fully tested and can easily be referenced in publications using the version number and the DOI.

The AtChem2 source code can be obtained in two ways:

1. with **git** (this is the best method for users who want to contribute to the model development, Chapt. 5):
   - ◇ Open the terminal. Move to the directory where AtChem2 will be installed.
   - ◇ Execute one of the following commands to copy the `master branch` to the local machine:
     - ○ if using HTTPS:
       `git clone https://github.com/AtChem/AtChem2.git`
     - ○ if using SSH:
       `git clone git@github.com:AtChem/AtChem2.git`
2. with the **archive file**:
   - ◇ Download the archive file of the *stable version* from the Releases page to the directory where AtChem2 will be installed.
   - ◇ Open the terminal. Move to the directory where the archive file was downloaded.
   - ◇ Execute `tar -zxvf *.tar.gz` or `unzip *.zip` (depending on which archive file was downloaded) to unpack the archive file.

The AtChem2 source code is now in a directory called `AtChem2` (if git was used) or `AtChem2-1.x` (if the archive file was downloaded). This directory is the AtChem2 *Main Directory*. In this manual, the *Main Directory* is assumed to be `$HOME/AtChem2/`, but it can be given any name and it can be located anywhere on the operating system according to the user's preferences.

2.3 Dependencies

AtChem2 has a number of dependencies (additional software tools and libraries): some are *required*, and without them the model cannot be installed or used, others are *optional*, and only needed to run the Testsuite (Sect. 5.2).

It is recommended to install all the dependencies into the same directory, called *Dependencies Directory*. In this manual, the *Dependencies Directory* is assumed to be `$HOME/AtChem-lib/`, but it can be given any name and it can be located anywhere on the operating system according to the user's preferences.

Before installing the dependencies, make sure that the requirements listed in Sect. 2.1 are met, and that the unix shell is *sh*-compatible (type `echo $SHELL` at the terminal to check the shell type).

2.3.1 Required dependencies

CVODE

AtChem2 uses the CVODE library, which is part of the open-source SUNDIALS suite [Hindmarsh et al., 2005], to solve the system of ordinary differential equations (ODE). The version of CVODE currently used in AtChem2 is v2.9.0 (part of SUNDIALS v2.7.0) and it can be installed using the script `install_cvode.sh` in the `tools/install/` directory.

1. Open the terminal. Move to the *Main Directory* (`cd ~/AtChem2/`).
2. If using a compiler other than `gcc`, open the `install_cvode.sh` script with a text editor and change the following line:

   ```
   -DCMAKE\_C\_COMPILER:FILEPATH=gcc \
   ```

3. Run the installation script (change the *Dependencies Directory* path as needed):

   ```
   ./tools/install/install_cvode.sh ~/AtChem-lib/
   ```

If the installation is successful, there is now a CVODE installation at `~/AtChem-lib/cvode/`. Take note of the CVODE library path, as it will be needed later to complete the configuration of AtChem2 (Sect. 2.4).

openlibm

openlibm is a portable implementation of the open-source mathematical library libm. Installing openlibm, and linking to it, allows reproducible results by ensuring the same implementation of several mathematical functions across platforms. The current version of openlibm can be installed using the script `install_openlibm.sh` in the `tools/install/` directory.

1. Open the terminal. Move to the *`Main Directory`* (`cd ~/AtChem2/`).
2. Run the installation script (change the *`Dependencies Directory`* path as needed):

```
./tools/install/install_openlibm.sh ~/AtChem-lib/
```

If the installation is successful, there is now an openlibm installation at `~/AtChem-lib/openlibm/`. Take note of the openlibm library path, as it will be needed later to complete the configuration of AtChem2 (Sect. 2.4).

Warning: the compilation of openlibm on **macOS** may fail with a message concerning the `dyld` library. This is analogous to the error that may occur the first time that AtChem2 is run on macOS and can be resolved in a similar way. See Sect. 2.4 for details.

2.3.2 Optional dependencies

numdiff

numdiff is an open-source tool to compare text files containing numerical fields. It is needed only to run the Testsuite, a battery of tests used during the development of AtChem2 to ensure that the model works properly and that changes to the source code do not result in unintended behaviour. Installation of numdiff is recommended for users who want to contribute to the development of AtChem2.

Use `which numdiff` to check whether the program is already installed on the operating system. If not, ask the system administrator. Alternatively, numdiff can be installed locally (e.g., in the *`Dependencies Directory`*) using the script `install_numdiff.sh` in the `tools/install/` directory.

1. Open the terminal. Move to the *`Main Directory`* (`cd ~/AtChem2/`).
2. Run the installation script (change the *`Dependencies Directory`* path as needed):

```
./tools/install/install_numdiff.sh ~/AtChem-lib/
```

3. Move to the `$HOME` directory (`cd ~`). Open the `.bash_profile` file (or the `.profile` file, depending on the system configuration) with a text

editor. Add the following line to the bottom of the file (change the *Dependencies Directory* path as needed):

```
export PATH=$PATH:$HOME/AtChem-lib/numdiff/bin
```

4. Close the terminal.

5. Reopen the terminal. Execute `which numdiff` to check that the program has been installed correctly.

FRUIT

FRUIT (FORTRAN Unit Test Framework) is an open-source testing library for Fortran. It requires a Ruby installation[2] and it is needed only to run the Unit tests, which are a subset of the Testsuite (Sect. 5.2). Installation of FRUIT is recommended only for users who want to contribute to the development of AtChem2.

The current version of FRUIT is 3.4.3 and it can be installed using the script `install_fruit.sh` in the `tools/install/` directory.

1. Open the terminal. Move to the `$HOME` directory (`cd ~`). Open the `.bash_profile` file (or the `.profile` file, depending on the system configuration) with a text editor. Add the following lines to the bottom of the file:

```
export GEM_HOME=$HOME/.gem
export PATH=$PATH:$GEM_HOME/bin
```

2. Close the terminal.

3. Reopen the terminal. Move to the *Main Directory* (`cd ~/AtChem2/`).

4. Run the installation script (change the *Dependencies Directory* path as needed):

```
./tools/install/install_fruit.sh ~/AtChem-lib/
```

If the installation is successful, there is now a FRUIT installation at `~/AtChem-lib/fruit_3.4.3/`. Take note of the FRUIT library path, as it will be needed later to complete the configuration of AtChem2 (Sect. 2.4).

[2]The current version of FRUIT does not compile with Ruby v3.2 due to some deprecated syntax. Ruby v3.0 or earlier must be used until the FRUIT developers release an update.

2.4 Install

After installation of the Dependencies, the procedure to install and configure AtChem2 is straightforward:

1. Open the terminal. Move to the *Main Directory* (`cd ~/AtChem2/`).

2. Copy the template `Makefile` from the `tools/install/` directory to the *Main Directory*:

   ```
   cp ./tools/install/Makefile.skel ./Makefile
   ```

3. Open the `Makefile` with a text editor. Set the variables `CVODELIBDIR`, `OPENLIBMDIR` and `FRUITDIR` to the paths of the CVODE, openlibm and FRUIT libraries, as indicated in Sect. 2.3. Use *full paths*, because relative paths may cause compilation errors. For example (change the *Dependencies Directory* path as needed):

   ```
   CVODELIBDIR = $(HOME)/AtChem-lib/cvode/lib
   OPENLIBMDIR = $(HOME)/AtChem-lib/openlibm
   FRUITDIR = $(HOME)/AtChem-lib/fruit_3.4.3
   ```

   If FRUIT has not been installed (it is optional and needed only to run certain tests), leave the default value for `$FRUITDIR`.

4. Compile AtChem2 using the build script (`build_atchem2.sh`) in the `build/` directory:

   ```
   ./build/build_atchem2.sh --mechanism=./model/mechanism.fac
   ```

   The build script creates an executable called `atchem2` in the *Main Directory*, using the default mechanism file `mechanism.fac` in the `model/` directory.

5. Execute `./atchem2` from the *Main Directory*. The command runs the model executable using the default chemical mechanism and settings, which are in the `model/` directory. The executable prints information to the terminal while it is running, such as the model settings, warning messages (if any), and timesteps. If the model run completes successfully, the terminal displays a message similar to the following:

   ```
   ------------------
    Final statistics
   ------------------
    No. steps = 546   No. f-s = 584   No. J-s = 912   No. LU-s = 56
    No. nonlinear iterations = 581
    No. nonlinear convergence failures = 0
    No. error test failures = 4

    Runtime = 0
    Deallocating memory.
   ```

When AtChem2 is run for the first time on a **macOS** system, it may abort with the following message:

```
dyld: Library not loaded: @rpath/libsundials_cvode.2.dylib
```

This error is related to the path of dynamic libraries on macOS: to resolve the issue, follow the suggestions on the wiki.

AtChem2 is now ready to be used. Optionally, the Testsuite can be run to check the model installation: go to Sect. 2.4.1 for more information. The directory structure of AtChem2 is described in Sect. 2.6 (see also Fig. 1.1). For detailed instructions on how to set up, configure, build and run the AtChem2 model, go to Chapts. 3 and 4.

2.4.1 Tests (optional)

The Testsuite can be used to verify that AtChem2 has been installed correctly and works as intended. This step is not necessary, but it is recommended for users who want to contribute to the development of AtChem2. Note that, in order to run the Testsuite, the optional dependencies have to be installed.

Open the terminal and execute one of the following commands from the *Main Directory*:

- ◇ `make alltests`: runs all the tests (requires numdiff and FRUIT).
- ◇ `make modeltests`: runs only the Model tests (requires numdiff).
- ◇ `make unittests`: runs only the Unit tests (requires FRUIT).

The command runs the requested tests, then prints the test output and a summary of the results to the terminal. The log files of the test runs can be found in the `tests/` directory.

2.5 Docker

As an alternative to the installation of AtChem2 described above, a containerized version of the model is available. Currently this is built only for AtChem2 v1.2.2 and is *experimental*; the following instructions are for guidance only.

The container is built on Rocky Linux 8.9 and pre-installs the CVODE and openlibm dependencies via the corresponding install scripts (Sect. 2.3.1). The image can be downloaded with the command:

```
docker pull ghcr.io/wacl-york/atchem2:1.2.2
```

When running the container, place the model files (e.g., configuration, constraints and mechanism files) in a folder that matches the AtChem2 directory structure. This folder is then mounted in the container with the name `/data_transfer/`. The chemical mechanism to be used is provided as a positional argument to the image (e.g., `./model/my_mech.fac`).

Example host file structure:

```
my_model_run
  |__ model
     |__ configuration
     |__ constraints
     |__ my_mech.fac
```

Example Docker run command:

```
docker run -it --rm -v /path/to/my_model_run:/data_transfer
       ghcr.io/wacl-york/atchem2:1.2.2 ./model/my_mech.fac
```

Output files will be copied to `my_model_run/model/output` on completion.

2.5.1 Running on Singularity/Apptainer

Some high-performance computing (HPC) systems use Singularity / Apptainer instead of Docker as their container engine. This image is compatible with those as well. The image can be converted to a `.sif` file via the command:

```
apptainer pull path/to/image/atchem2.sif
          docker://ghcr.io/wacl-york/atchem2:1.2.2
```

Example Apptainer run command:

```
apptainer run --bind /path/to/my_model_run/:/data_transfer/
          path/to/image/atchem2.sif ./model/my_mech.fac
```

Note that the leading `/` is important when mounting the volume for apptainer, as the container opens in the user's home directory, whereas Docker opens at `/`.

2.6 Model Structure

AtChem2 is organized in several directories which contain the source code, the compilation files, the chemical mechanism, the model configuration and output, several scripts and utilities, and the Testsuite.

The directory structure of AtChem2 is derived from the directory structure of AtChem-online, but it was substantially changed with the release of

Table 2.1: Directory structure of AtChem2.
(§) Not present in AtChem-online. (†) Since version 1.2.2. (‡) In `model/configuration` before version 1.3.

Version 1.0 and earlier	Version 1.1 and later	Description
–	`build/`	scripts to build the model
–	`doc/`	user manual and other documents
–	`docker/` (†)	Docker-related files
–	`mcm/`	MCM data files and example mechanism files
`modelConfiguration/`	`model/configuration/`	chemical mechanism, model and solver configuration files
`speciesConstraints/`	`model/constraints/species/`	model constraint files: chemical species
`environmentConstraints/`	`model/constraints/environment/`	model constraint files: environment variables
`environmentConstraints/`	`model/constraints/photolysis/`	model constraint files: photolysis rates, JFAC
–	`model/sharedlib/` (‡)	pre-compiled mechanism shared library
`modelOutput/`	`model/output/`	model results: chemical species, environment variables, photolysis rates, production/loss rates of selected species and diagnostic variables
`instantaneousRates/`	`model/output/reactionRates/`	model results: reaction rates of every reaction in the chemical mechanism
`obj/`	`obj/`	module and object files generated by the Fortran compiler
`src/`	`src/`	Fortran source files
`travis/` (§)	`tests/` (†)	Testsuite scripts, logs and files
`tools/`	`tools/`	installation scripts, plotting tools and other utilities

version 1.1 (November 2018). Tab. 2.1 shows the new directory structure and, for reference, the original one.

In AtChem2 v1.1 (and later versions), the directories `build/`, `mcm/`, `obj/` and `src/` contain the build scripts, the MCM data files, the files generated by the Fortran compiler, and the Fortran source code, respectively. The directory `tests/` contains the files and scripts necessary to run the Testsuite; the directory `docker/` contains the scripts necessary to run AtChem2 as a Docker container (since v1.2.2).

For the majority of the users, the most important directories are `doc/`, which contains the user manual; `tools/`, which contains the installation and plotting scripts; `model/`, which contains the model configuration, input, output and (usually) the chemical mechanism. Detailed information on these three directories can be found in Sects. 2.6.1 and 2.6.2, respectively.

2.6.1 The `doc/` and `tools/` directories

The `doc/` directory contains the AtChem2 user manual (this document, `AtChem2-Manual.pdf`), along with the corresponding LaTeX files and figures. It includes the poster presented at the 2018 Atmospheric Chemical Mechanisms Conference [Sommariva et al., 2018], which describes the architecture and features of AtChem2 and AtChem-online (`AtChem_poster_ACM2018.pdf`).

The `tools/` directory contains the script used to update the AtChem2 version number in each development cycle (`update_version_number.sh`), two scripts to check the consistency of the Fortran code (`fix_style.py` and `fix_indent.py`, see Sect. 5.3), and the following subdirectories:

- ◇ `install/`: contains the template `Makefile` (`Makefile.skel`) and the scripts to install the Dependencies.
- ◇ `plot/`: contains some basic scripts, in various programming languages, to plot the model output.

In earlier versions of AtChem2, the `tools/` directory also included the build scripts, which have been moved to the `build/` directory with the release of version 1.2.

The plotting scripts in `tools/plot/` are very simple and produce identical outputs. They are intended only to give the user a quick overview of the model results for validation and diagnostic purposes. It is recommended to use a proper data analysis software package (e.g., R, Python, Igor, Origin, etc.) to process and analyze the model results.

The plotting scripts are written in different programming languages: gnuplot, Octave[3], Python, R. One or more of these environments is probably already installed on the operating system: check the local documentation or ask the system administrator. All plotting scripts require one argument – the path of the directory containing the model output – and produce the same output: one file called `atchem2_output.pdf` in the model output directory.

To run a plotting script, open the terminal and execute one of the following commands from the *`Main Directory`* (change the path of the model output directory as needed):

- ◇ `gnuplot -c ./tools/plot/plot-atchem2.gp ./model/output/`
- ◇ `octave ./tools/plot/plot-atchem2.m ./model/output/`
- ◇ `python ./tools/plot/plot-atchem2-numpy.py ./model/output/`
- ◇ `python ./tools/plot/plot-atchem2-pandas.py ./model/output/`
- ◇ `Rscript --vanilla ./tools/plot/plot-atchem2.r ./model/output/`

[3]GNU Octave is an open-source implementation of MATLAB. The script `plot-atchem2.m` works with both Octave and MATLAB.

2.6.2 The `model/` directory

The `model/` directory is the most important from the point of view of the user. It is referred to as the *Model Directory* and it contains all the information required to set up and run an AtChem2 box-model, together with the results of the model run.

The *Model Directory* is a special type of directory: first, there can be more than one (with different names). Second, although by default the *Model Directory* is located inside the *Main Directory* and is called `model/`, it can be located anywhere the user wants and it can be given any name. The user only needs to provide the path of the *Model Directory* as an argument to the build script (`build/build_atchem2.sh`) and to the `atchem2` executable, as explained in Sects. 4.3 and 4.4.

The *Model Directory* contains a number of subdirectories:

- ◇ *Configuration Directory* for the model configuration files.
  By default, it is `model/configuration/`.
- ◇ *Constraints Directory* for the model constraint files.
  By default, it is `model/constraints/`.
- ◇ *Shared Library Directory* for the pre-compiled chemical mechanism shared library (`mechanism.so`).
  By default, it is `model/sharedlib/`.
- ◇ *Output Directory* for the model output.
  By default, it is `model/output/`.

Note that, although the chemical mechanism file (`.fac` or `.kpp`) does not need to be located inside the *Model Directory*, it is good practice to do so in order to keep all the model setup information together.

This approach gives the user the flexibility to run multiple versions of the same model (e.g., with different configurations and/or chemical mechanisms) or multiple models (e.g., for separate projects) at the same time, without having to recompile the source code and create a different executable every time. Batch model runs are therefore easy to do, since all the components of the model that have to be modified for each run are contained in the same directory. For further information, see Sect. 3.1.5.

Chapter 3

Model Setup

3.1 Chemical Mechanism

The chemical mechanism is the core of an atmospheric chemistry model. In AtChem2, the chemical mechanism file is written in **FACSIMILE format** and has the extension `.fac`. The format is used to describe chemical reactions in the commercial FACSIMILE Kinetic Modelling Software. Historically, this software and the format have been closely associated with the Master Chemical Mechanism. The extraction tool on the MCM website generates `.fac` files that can be used directly in AtChem2 (Sect. 3.1.3).

Since version 1.2.3, AtChem2 also supports the **KPP format** for chemical mechanisms. The `.kpp` mechanism file, which should have a similar format to the KPP files generated by the MCM extraction tool, is automatically converted to a `.fac` file during the build process (Sect. 4.1.1).

3.1.1 FACSIMILE format

Chemical reactions are described in FACSIMILE format using the notation:

```
% k : A + B = C + D ;
```

where `k` is the rate coefficient, `A` and `B` are the reactants, `C` and `D` are the products. The definition of a chemical reaction in FACSIMILE starts with the `%` character and ends with the `;` character. The `:` character separates the rate coefficient from the reactants and products.

Since version 1.2.3, AtChem2 accepts stoichiometric coefficients in reaction definitions. The coefficients must be written before a species name, with or without a whitespace in between; they can be expressed as integer or decimal values. For example, the following reaction definitions are all equivalent:

```
% k : A + A = B + B + C ;
% k : 2A = 2B + C ;
```

```
% k : 2.0A = 2.0B + C ;
% k : 2 A = 2 B + C ;
% k : 2.0 A = 2.0 B + C ;
% k : A + A = 2B + 1.0 C ;
```

Note that the names of the rate coefficients and of the chemical species (products and reactants) must begin with a letter and not with a numerical character, although numbers may occur at other points of the variable name. This is to avoid confusion with the stoichiometric coefficients.

The rate coefficient of a chemical reaction (`k`) can be a constant number or, more commonly, can be calculated as a function of temperature (`TEMP`), air density (`M`), water vapour (`H2O`), and other environment variables (Sect. 3.4).

Comments can be inserted in the `.fac` file to document and annotate the chemical mechanism: in FACSIMILE format, comments are enclosed between the `*` and `;` characters, and are ignored by the build scripts. A basic chemical mechanism, with comments and calculated rate coefficients, has the following format:

```
* Tropospheric O3-NOx cycle ;
* Kinetic data from Atkinson et al., ACP, 2004 ;
*;
% J_NO2                        : NO2 = NO + O ;
% 5.6D-34*M*(TEMP/300)@-2.6    : O + O2 = O3 ;
% 1.4D-12*EXP(-1310/TEMP)      : NO + O3 = NO2 + O2 ;
```

In the example above, the photolysis rate of NO_2 (`J_NO2`) can be calculated by AtChem2 as a function of latitude, longitude and solar zenith angle, or can be provided by the user (Sect. 3.5).

Complex mathematical expressions can be used to calculate the rate coefficients, in which case they have to be defined before the chemical reactions that use them (typically, these are combination or dissociation reactions). For example:

```
* Formation of nitric acid (HNO3) in the gas-phase ;
*;
* Rate coefficient (Atkinson et al., ACP, 2004) ;
K80 = 3.3D-30*M*(TEMP/300)@-3.0 ;
K8I = 4.1D-11 ;
KR8 = K80/K8I ;
FC8 = 0.4 ;
NC8 = 0.75-1.27*(LOG10(FC8)) ;
F8 = 10@(LOG10(FC8)/(1+(LOG10(KR8)/NC8)**2)) ;
KMT08 = (K80*K8I)*F8/(K80+K8I) ;
*;
* Chemical Reaction ;
% KMT08 : OH + NO2 = HNO3 ;
```

In FACSIMILE format, chemical reactions can also be written without reactants or products. This feature can be used to implement simple descriptions of non-chemical processes in a box-model. For example, dilution (Sect. 3.4.8), surface deposition and direct emission of a chemical species can be parametrized as:

```
* Deposition velocity of O3 = 1.4 cm s-1 ;
* Boundary layer height (BLHEIGHT) in cm ;
% 1.4/BLHEIGHT  :  O3 =  ;
*;
* Emission rate of NO2 = 1e8 molecule cm-3 s-1 ;
% 1D+8  :  = NO2  ;
```

More sophisticated approaches to describe non-chemical processes can be implemented by defining complex mathematical expressions to calculate the corresponding "rate coefficients", as explained above. Alternatively, since version 1.2.3, AtChem2 allows the user to create custom Fortran functions: for detailed instructions, go to Sect. 3.1.4.

During the build process, the chemical mechanism file has to be parsed by the `mech_converter.py` script, which expects the chemical mechanism file to have four sections:

Generic rate coefficients : contains the equations of the generic rate coefficients, typically used when experimental or theoretical kinetic data are not available.

Complex reactions : contains the equations of the complex rate coefficients (e.g., for combination/dissociation reactions or parametrized non-chemical processes).

Peroxy radicals : contains the calculation of the sum of organic peroxy radicals (RO_2) – go to Sect. 3.1.2 for details.

Reaction definitions : contains the definitions of the chemical reactions and of the parametrized non-chemical processes, such as aerosol uptake, dry deposition, etc.

For the `mech_converter.py` script to work, the beginning of each section must be delimited by a header consisting of a single comment line. The header must always be present, even though the corresponding section can be empty (e.g., if a mechanism other than the MCM is used). Examples of minimal chemical mechanism files (`mechanism_skel.*`) can be found in the `mcm/` directory for reference. In addition, a chemical mechanism file downloaded from the MCM website (`mechanism.fac`) is included in the `model/` directory.

3.1.2 RO2 sum

The sum of organic peroxy radicals (RO_2) is a key feature of the Master Chemical Mechanism. Organic peroxy radicals react with HO_2, with themselves and with other organic peroxy radicals: given that there are over 1000 RO_2 in the MCM, the number of possible self- and cross-reactions of these species is of the order of 10^6, which presents a significant computational challenge. The RO2 sum is used in the MCM to reduce the number of peroxy radical permutation reactions. More information can be found in the MCM protocol papers [Jenkin et al., 1997, Saunders et al., 2003].

AtChem2 is primarily designed to run box-models based upon the Master Chemical Mechanism, which requires the `.fac` file to include a **Peroxy radicals** section, as explained in Sect. 3.1.1. This section must have a header and has the following format:

```
* Peroxy radicals. ;
RO2 = RO2a + RO2b + RO2c + ... ;
```

where `RO2a`, `RO2b`, `RO2c`, are the organic peroxy radicals in the chemical mechanism. If there are no organic peroxy radicals in the chemical mechanism (or if the mechanism is not based upon the MCM), the RO2 sum section must still be present in the `.fac` file, but it is left empty:

```
* Peroxy radicals. ;
RO2 = ;
```

The RO2 sum is automatically generated from the `.fac` file during the build process. The script `mech_converter.py` compares the list of species in the **Peroxy radicals** section of the `.fac` file with the list of RO_2 extracted from the MCM database, which is included in the `mcm/` directory. By default, AtChem2 uses the MCM v3.3.1 and the corresponding list of organic peroxy radicals is in the `peroxy-radicals_v3.3.1` file. Lists of organic peroxy radicals for other versions of the MCM are included in the `mcm/` directory and can be used by changing the MCM version using the build script flag `--mcm`, as explained in Sect. 4.3.

If one or more chemical species are included in the **Peroxy radicals** section of the `.fac` file but are not present in the list of RO_2 extracted from the MCM database, the `mech_converter.py` script prints a warning message to the terminal:

```
      !!! WARNING !!!
The following species are not present in the RO2 reference list:
  speciesX speciesY
Should they be included in the RO2 sum?
```

In this case, the user must check manually whether the species in question are organic peroxy radicals, amend the **Peroxy radicals** section of the `.fac` file accordingly, and rerun the `build_atchem2.sh` script.

It is important to ensure that the RO2 sum is accurate, because many reactions in the MCM depend on this parameter. This means that all the organic peroxy radicals in the chemical mechanism must be included in the RO2 sum, and that the RO2 sum must include only organic peroxy radicals[1]. The model calculates the RO2 sum at runtime and, by default, outputs the value of this variable to the `environmentVariables.output` file (Sect. 4.5).

3.1.3 MCM extraction

The MCM website provides a convenient tool to download the whole Master Chemical Mechanism, or subsets of it, in FACSIMILE format. A brief overview of the process is given here; more information can be found on the MCM website.

First, select the species of interest using the MCM browser and add them to the "Marklist". Then proceed to the MCM export tool and select the option "FACSIMILE" as output format. Make sure to also select the following options, so that all the required headers (Sect. 3.1.1) are included in the generated `.fac` file:

```
[x] Include inorganic reactions?
[x] Include generic rate coefficients?
```

Click on the "Download" button and save the `.fac` file in a directory of choice – such as the `model/` directory (Sect. 2.6.2). The procedure to generate the chemical mechanism file is identical if a `.kpp` file is desired (just select the option "KPP" instead of "FACSIMILE" as output format). The chemical mechanism file downloaded from the MCM website is a simple ASCII text file and is ready to be used in AtChem2. If modifications are required (e.g., chemical reactions to be added, deleted or modified), the user can open the file with a text editor and modify the chemical mechanism as needed.

3.1.4 Custom functions

The Fortran file `customRateFuncs.f90` allows the user to define custom functions that can be called from within the chemical mechanism to enhance the capability of the model[2]. These functions can be used to parametrize heterogeneous reactions, aerosol formation, surface deposition, emissions and other non-chemical processes, as discussed in Sect. 3.1.1.

[1] The hydroperoxyl radical (HO_2) is a peroxy radical but is not an organic molecule, and therefore it *must not* be included in the RO2 sum.

[2] This feature is available only in version 1.2.3 or later.

The Fortran file `customRateFuncs.f90` is located in the *`Configuration Directory`* (by default, `model/configuration/`, see Sect. 3.6) and contains an empty Fortran module which can be modified by the user to add any number of custom functions. During the build process, the `mech_converter.py` script identifies the names of all the custom functions in the module and includes them into the file `mechanism.f90` (Sect. 3.1.5). If no custom functions have been defined, the content of the module is ignored.

Note that the custom functions must be defined as Fortran functions and not as subroutines, that is, the Fortran keyword "function" must be used in the declaration. Environment variables (Sect. 3.4), such as temperature and pressure, can be used as arguments in a custom function. Below is an example of a modified `customRateFuncs.f90` file which defines a basic function to multiply two numbers:

```
module custom_functions_mod
  implicit none

contains

  ! Function to multiply two numbers
  pure function multiplyNumbers( inNum1, inNum2 ) result ( outNum )
    real, intent(in) :: inNum2, inNum1
    real :: outNum

    outNum = inNum1 * inNum2

    return
  end function multiplyNumbers

end module custom_functions_mod
```

The "multiplyNumbers()" function can then be called in the `.fac` file, together with the regular FACSIMILE notation (Sect. 3.1.1). For example:

```
% multiplyNumbers(2.0, 0.5) : A = B ;
```

It is the user's responsibility to ensure that the custom Fortran code is correct. Although optional, it is recommended to use the Python scripts `fix_style.py` and `fix_indent.py` in the `tools/` directory to check that the Fortran code follows the AtChem2 coding guidelines, which are described in Sect. 5.3.

3.1.5 Build process

AtChem2 is built using the scripts in the `build/` directory. In this section, the build process is briefly outlined; more detailed instructions can be found in Sect. 4.3.

The Python script `mech_converter.py` – called by the `build_atchem2.sh` script – converts the chemical mechanism into a Fortran-compatible format by generating a number of files:

- ◇ `mechanism.f90` contains the Fortran equations to calculate the rate coefficients of each reaction in the chemical mechanism.
- ◇ `mechanism.o` is the compiled Fortran code and `mechanism.so` is the mechanism shared library, i.e. the pre-compiled chemical mechanism.
- ◇ `mechanism.species` contains the chemical species in the chemical mechanism. The file has no header. The first column is the ID number of the species; the second column is the name of the species. For example:

```
1 O
2 O3
3 NO
4 NO2
```

- ◇ `mechanism.reac` and `mechanism.prod` contain the reactants and the products (respectively) of each reaction in the chemical mechanism. The files have a one-line header with the total number of species and reactions in the mechanism, and the total number of equations in the **Generic rate coefficients** and **Complex reactions** sections of the `.fac` file (Sect. 3.1.1). The first column is the ID number of the reaction; the second column is the ID number of the species (from `mechanism.species`) which is a reactant or a product in that reaction; the third column is the stoichiometric coefficient of the reaction. For example:

```
29 71 139 numberOfSpecies numberOfReactions numberOfGenericComplex
1 1 1.0
2 1 1.0
3 1 1.0
3 2 1.0
```

- ◇ `mechanism.ro2` contains the organic peroxy radicals (RO_2) in the chemical mechanism. The file has a one-line header as a Fortran comment. The first column is the ID number of the peroxy radical (from `mechanism.species`); the second column is the name of the peroxy radical as a Fortran comment. For example:

```
! Note that this file is automatically generated by
  build/mech_converter.py -- Any manual edits to this file
  will be overwritten when calling build/mech_converter.py
23 !CH3O2
26 !C2H5O2
```

```
28 !IC3H7O2
29 !NC3H7O2
```

The files generated during the build process are saved in the *Shared Library Directory* which, by default, is `model/sharedlib/`. The location can be changed using the build script flag `--shared_lib`, as explained in Sect. 4.3 (see also Sect. 2.6.2).

Important Note: all the files in the *Shared Library Directory* are automatically generated and overwritten every time the build script is executed. Users should not modify any of these files.

3.2 Model Parameters

The model parameters control the general setup of the model, such as the runtime, the interpolation methods, the output frequencies, etc. The model parameters, which are described below, are set in the `model.parameters` file in the *Configuration Directory*.

- ◇ **number of steps**, **step size**. The model runtime is determined by the number of steps and by the step size (in seconds). The step size controls the frequency of the model output for the chemical species listed in `outputSpecies.config` (Sect. 3.6.4), as well as for the environment variables, the photolysis rates and the diagnostic variables. Note that the step size is *distinct* from the integration step, which is independently controlled by CVODE.
  For example, a model run of 2 hours, with output every 5 minutes, requires 24 steps with a step size of 300 seconds: $24 \times 300 = 7200$ seconds = 2 hours

- ◇ **species interpolation method**, **conditions interpolation method**. Interpolation method used for the constrained chemical species, and for the constrained environment variables and the photolysis rates, respectively. Two interpolation methods are currently implemented in AtChem2: piecewise constant and piecewise linear (Sect. 4.2.3). The default option is `2` (piecewise linear interpolation).

- ◇ **rates output step size**. Frequency (in seconds) of the model output for the production and loss rates of selected species – listed in `outputRates.config` (Sect. 3.6.3). This is used for the rate of production and destruction analysis (ROPA/RODA, Sect. 4.5.1).

- ◇ **model start time**. Start time of the model (in seconds) calculated from midnight of the **day**, **month**, **year** parameters – see below.
  For example, a start time of 3600 means the model run starts at 1:00

in the morning and a start time of 46800 means the model run starts at 1:00 in the afternoon (13:00). The **model stop time** does not need to be specified, and is automatically calculated by the model as:

$$\text{stoptime} = \text{starttime} + (\text{numberofsteps} \times \text{stepsize}) \tag{3.1}$$

Note that when one or more variables are constrained, the time interval between the model start time and stop time *must be* equal to, or less than, the time interval of the constrained data – go to Sect. 4.2.3 for more information.

- ◇ **jacobian output step size**. Frequency (in seconds) of the model output for the Jacobian matrix. If this parameter is set to `0` (default option), the Jacobian is not saved. The `jacobian.output` file generated by the model can be very large, especially if the chemical mechanism has many reactions and/or the model runtime is long. Therefore, it is recommended to output the Jacobian matrix only if needed.

- ◇ **latitude** and **longitude**. Geographical coordinates (in degrees). Latitude North is positive and latitude South is negative; longitude East is negative and longitude West is positive[3]. Latitude and longitude are used to calculate the solar angles, which are needed for the calculation of the photolysis rates (Sect. 3.5.3).

- ◇ **day**, **month** and **year**. Start date of the model simulation. The model time is in UTC (i.e. GMT timezone) and is calculated in seconds since midnight of the start date.

- ◇ **reaction rates output step size**. Frequency (in seconds) of the model output for the reaction rates of every reaction in the chemical mechanism. By default, the reaction rates are saved in the subdirectory `reactionRates/` of the *`Output Directory`* (Sect. 4.5). Note that this parameter is different from **rates output step size** (see above), which sets the output frequency for the production and loss rates of a limited number of species of interest.

3.3 Solver Parameters

The solver parameters control the behaviour of the ordinary differential equations (ODE) solver and are set in the `solver.parameters` file in the *`Configuration Directory`*. The complete explanation of the solver parameters can be found in the documentation of CVODE.

[3]The standard geographical convention is that longitude East is positive and longitude West is negative. AtChem2 uses the opposite convention for reasons of backward compatibility. This behaviour may change in future versions.

- ◇ **atol** (positive real) and **rtol** (positive real): absolute and relative tolerance values for the solver.
- ◇ **delta main** (positive real): linear convergence tolerance factor of the GMRES linear solver.
- ◇ **lookback** (positive integer): maximum Krylov subspace dimension of the GMRES linear solver.
- ◇ **maximum solver step size** (positive real): maximum size of the timesteps that the solver is allowed to use (in seconds).
- ◇ **maximum number of steps in solver** (positive integer): maximum number of steps used by the solver before reaching `tout`, i.e. the next output time.
- ◇ **solver type** (integer): selection of the linear solver to use: `1` for GMRES, `2` for GMRES preconditioned with a banded preconditioner (default option), `3` for a dense solver.
- ◇ **banded preconditioner upper bandwidth** (integer): only used when `solver type = 2`.
- ◇ **banded preconditioner lower bandwidth** (integer): only used when `solver type = 2`.

3.4 Environment Variables

The environment variables define the physical parameters of the model, such as temperature, pressure, humidity, solar angles, boundary layer height, etc. These variables are set in the `environmentVariables.config` file in the *`Configuration Directory`*.

The environment variables can have a fixed value, i.e. a numerical or logical constant, or can be constrained to a timeseries of measured values (`CONSTRAINED`), depending on their type. If a variable is constrained, the corresponding data file must be present in the subdirectory `environment/` of the *`Constraints Directory`* (Sect. 4.2.1). Some environment variables can be calculated by the model (`CALC`) and some can be deactivated if they are not needed (`NOTUSED`).

By default, the environment variables are either set to `NOTUSED` or to a fixed value. The environment variables used in AtChem2 are described below, together with their possible settings, units, and default values.

3.4.1 TEMP

Ambient temperature (in K).

- ◇ fixed value
  ↪ default: 288.15
- ◇ constrained

3.4.2 PRESS

Ambient pressure (in mbar).

- ◇ fixed value
  ↪ default: 1013.25
- ◇ constrained

3.4.3 RH

Relative humidity (in %). It is required only if H2O (see below) is set to `CALC`, otherwise it must be set to `NOTUSED`.

- ◇ fixed value
- ◇ constrained
- ◇ not used
  ↪ default: `NOTUSED` (−1)

3.4.4 H2O

Water concentration (in molecule cm^{-3}). If `H2O` is set to `CALC`, then RH (see above) must be set to a fixed value or `CONSTRAINED`.

- ◇ fixed value
  ↪ default: 4.45e+17
- ◇ constrained
- ◇ calculated

3.4.5 DEC

Sun declination (in radians) is the angle between the center of the Sun and Earth's equatorial plane.

- ◇ fixed value
- ◇ constrained
- ◇ calculated
  ↪ default: `CALC`

3.4.6 BLHEIGHT

Boundary layer height. It is required only if the model includes non-chemical processes such as the emission or deposition of chemical species. The unit is centimetre (cm) or metre (m), depending on how these processes are parametrized in the chemical mechanism – go to Sect. 3.1.1 for details.

- ◇ fixed value
- ◇ constrained
- ◇ not used
  ↪ default: `NOTUSED` (−1)

3.4.7 ASA

Aerosol surface area. It is required only if the model includes heterogeneous chemical reactions. The unit is area (e.g., cm^2, μm^2, nm^2) per volume (e.g., m^3, cm^3), depending on how these reactions are implemented in the chemical mechanism.

- ◇ fixed value
- ◇ constrained
- ◇ not used
  ↪ default: `NOTUSED` (−1)

3.4.8 DILUTE

Dilution rate (in s^{-1}). It is required only if the model includes the dilution of chemical species.

- ◇ fixed value
- ◇ constrained
- ◇ not used
  ↪ default: `NOTUSED` (−1)

When `DILUTE` is set to a fixed value or constrained, the chemical mechanism is automatically modified during the build process: for each species in the chemical mechanism, a "loss reaction" is added to the mechanism with a "rate coefficient" equal to the dilution rate[4]. For example, the following "reactions" are added to the "Tropospheric O3-NOx cycle" mechanism shown in Sect. 3.1.1:

[4]The behaviour of `DILUTE` was changed with version 1.2. Previously, the user had to add manually to the chemical mechanism a "reaction" using `DILUTE` for each species for which dilution was required. The new approach makes the use of `DILUTE` both more intuitive and more accurate.

```
% DILUTE : NO2 = ;
% DILUTE : NO = ;
% DILUTE : O = ;
% DILUTE : O3 = ;
```

Note that AtChem2 treats molecular oxygen (O_2) and nitrogen (N_2) as model parameters rather than chemical species. Their concentrations are calculated by the model as a function of temperature and pressure and, therefore, `DILUTE` is not applied to O_2 and N_2.

3.4.9 JFAC

Correction factor used to adjust the photolysis rates – e.g., to account for the effect of clouds. `JFAC` can have a value between `0` (photolysis rates are set to zero) and `1` (photolysis rates are not corrected). `JFAC` is not applied to constant or constrained photolysis rates (Fig. 3.1). For detailed information on this parameter, go to Sect. 3.5.4.

- ◇ fixed value
  ↪ default: 1
- ◇ constrained
- ◇ calculated

3.4.10 ROOF

Flag to switch the photolysis rates on/off. It is needed mostly for simulations of some environmental chamber experiments, where the roof of the chamber can be opened/closed or the lamps can be turned on/off. When `ROOF` is set to `CLOSED`, all the photolysis rates are set to zero, including those that are constant or constrained. This is different from setting `JFAC` to `0`, which only applies to the calculated photolysis rates (Fig. 3.1).

The parameter `ROOF` can be set only to `OPEN` (default) or `CLOSED` and cannot be constrained.

3.5 Photolysis Rates

The photolysis rates are the rate coefficients (in s^{-1}) of the photolysis reactions; they are indicated in FACSIMILE format with the notation `J<i>` – where `i` is the ID number assigned by the MCM to each photolysis reaction (or to a group of photolysis reactions, as per MCM protocol). AtChem2 implements the MCM parametrization (Sect. 3.5.3) to calculate the photolysis rates; for a comparison between calculated and measured photolysis rates see Sommariva et al. [2020]. Alternatively, the photolysis rates can be set to a constant value or constrained to measured data. The following rules apply:

1. If a photolysis rate is set to a constant value, it assumes the given value; the other photolysis rates are set to zero.

2. If a photolysis rate is constrained, it assumes the values in the corresponding constraint file; the other photolysis rates are calculated.

3. If no photolysis rate is set to constant or constrained, the model calculates all photolysis rates.

Fig. 3.1 shows how AtChem2 combines constant, constrained and calculated photolysis rates, as well as the use of the photolysis correction factor `JFAC` (Sect. 3.5.4). In addition, the environment variable `ROOF` can be used to turn the photolysis rates on or off, as explained in Sect. 3.4.10.

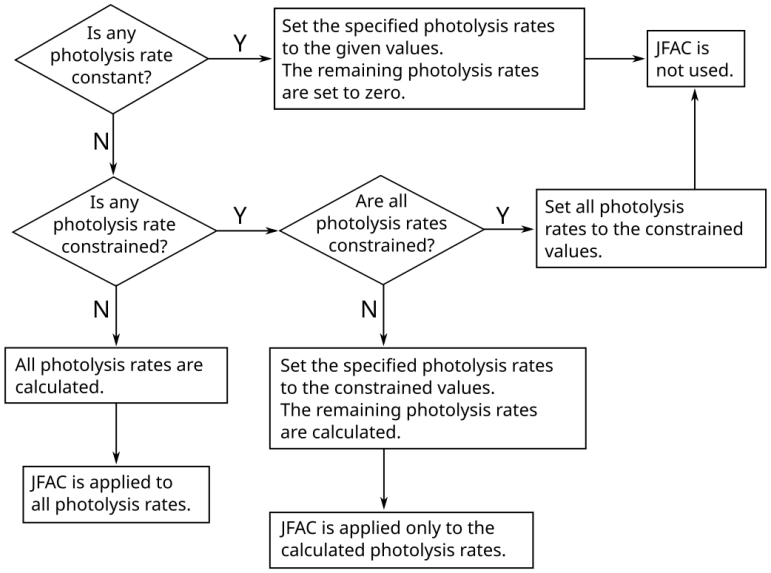


Figure 3.1: Photolysis rates and `JFAC` in AtChem2.

3.5.1 Constant photolysis rates

The typical scenario for constant photolysis rates is a lamp or solar simulator in an environmental chamber. All the photolysis rates in the chemical mechanism must be given a fixed value in the `photolysisConstant.config` file, otherwise they are automatically set to zero (Fig. 3.1). This approach allows the user to model individual photolytic processes and/or to account for lamps that emit only in certain spectral windows.

The format of `photolysisConstant.config` is described in Sect. 3.6.5. If the file is empty, the photolysis rates are constrained or calculated by the model as explained below.

3.5.2 Constrained photolysis rates

All photolysis rates can be constrained to measured values. In this case, the name of the constrained photolysis rate (e.g., `J2`) must be listed in the `photolysisConstrained.config` file (Sect. 3.6.6); a corresponding file with the constraint data must be present in the subdirectory `photolysis/` of the *`Constraints Directory`* (see Sect. 4.2.1 for details).

It is not always possible to measure – and therefore to constrain – all the required photolysis rates. The photolysis rates that are not constrained (i.e. not listed in `photolysisConstrained.config`) are calculated by the model using the MCM parametrization, described in the next section.

3.5.3 Calculated photolysis rates

AtChem2 implements the parametrization of photolysis rates used by the Master Chemical Mechanism, which is described in detail in the MCM protocol papers [Jenkin et al., 1997, Saunders et al., 2003]. Briefly, the photolysis rate (`J`) is calculated as a function of the solar zenith angle according to the following equation:

$$J = l \times (cosX)^m \times e^{(-n \times secX)} \times \tau \tag{3.2}$$

where l, m, n are empirical parameters; $cosX$ is the cosine of the solar zenith angle; $secX$ is the inverse of $cosX$ (i.e. $secX = 1/cosX$); τ is the transmission factor. The empirical parameters l, m, n are different for each version of the MCM: by default, AtChem2 uses the set of parameters of version 3.3.1 (from file `mcm/photolysis-rates_v3.3.1`), but it is possible to use the parameters of other versions via the build script flag `--mcm`, as explained in Sect. 4.3.

The transmission factor τ accounts for the loss of natural or artificial light in some environmental chambers: the default value of τ is 1, which means there is perfect transmittance of light through the chamber walls or windows. The value of τ can be changed: see the file `mcm/INFO.md` for instructions. More information can be found on the photolysis page of the MCM website.

The solar zenith angle (SZA) is the angle between the local vertical (zenith) and the center of the Sun. The SZA is measured in radians and is calculated by the model from the following variables: latitude and longitude (both set in the `model.parameters` file, Sect. 3.2); Sun declination (`DEC`, Sect. 3.4.5) – an environment variable which can be constrained, set to a constant value or calculated by the model; time of the day, which is calculated based on the start date and the model time. The equations to calculate the solar zenith angle and the Sun declination are described in Madronich [1993].

3.5.4 JFAC calculation

Measurements of ambient photolysis rates typically show short-term variability due to changing meteorological conditions, such as clouds and aerosol [Sommariva et al., 2020]. This information is retained in the constrained photolysis rates, but it is not included in the calculated ones. To account for the ambient variability, the calculated photolysis rates can be scaled by a constant or time-dependent correction factor – the environment variable `JFAC` (Sect. 3.4.9) – defined as:

$$\text{JFAC} = \frac{j_{meas}}{j_{calc}} \tag{3.3}$$

where j_{meas} and j_{calc} are the measured and calculated (with the MCM parametrization, Sect. 3.5.3) photolysis rates of a reference species. NO_2 is often used as reference because $j(NO_2)$ is one of the most frequently measured photolysis rates, in which case:

$$\text{JFAC} = \frac{j(NO_2)}{\text{J4}} \tag{3.4}$$

The parameter `JFAC` is by default set to 1, which means that the calculated photolysis rates are not scaled; `JFAC` can be set to any value between 0 and 1 (Sect. 3.4.9), or it can be constrained (Sect. 4.2.1). Note that only the photolysis rates calculated with the MCM parametrization are scaled by `JFAC`: the constrained and the constant photolysis rates are never scaled (Fig. 3.1).

AtChem2 can calculate `JFAC` automatically at runtime. To use this option, edit the `environmentVariables.config` file (Sect. 3.6.1) and set `JFAC` to the name of the photolysis rate used as reference (e.g., `J4`). The reference photolysis rate must also be constrained – i.e. it must be listed in the `photolysisConstrained.config` file (Sect. 3.6.6) and the corresponding constraint file must be present in the `model/constraints/photolysis/` directory.

Warning: currently, the calculation of `JFAC` at runtime does not work well[5], especially in situations when the reference photolysis rate is highly variable, and at sunrise/sunset. Therefore, it is recommended to calculate `JFAC` offline and then to constrain it (go to Sect. 4.2 for details).

3.6 Configuration Files

The configuration files contain the settings of the environment variables, the chemical species, the photolysis rates, as well as the model constraints and the model output. All the configuration files have the extension `.config`

[5] See issue #16.

and are located in the *Configuration Directory* which, by default, is `model/configuration/`. The location can be changed using the build script flag `--configuration`, as explained in Sect. 4.3 (see also Sect. 2.6.2). The content and the format of each `.config` file are described below.

The *Configuration Directory* also contains the files with the settings of the model (`model.parameters`) and of the solver (`solver.parameters`) – described in Sect. 3.2 and 3.3, respectively – as well as the custom Fortran functions that can be defined by the user (`customRateFuncs.f90`, see Sect. 3.1.4).

3.6.1 environmentVariables.config

This file specifies the settings of the environment variables, i.e. the physical parameters of the model. The file has three columns: the first two are the ID number and the name of the environment variable. The third column – the only one that should be modified by the user – contains the corresponding value. The valid settings and the units of each environment variable are detailed in Sect. 3.4. For example:

```
1   TEMP        293
2   PRESS       1013
3   RH          CONSTRAINED
4   H2O         CALC
5   DEC         CALC
6   BLHEIGHT    8e+4
7   DILUTE      NOTUSED
8   JFAC        CONSTRAINED
9   ROOF        OPEN
10  ASA         2e-5
```

If an environment variable is constrained, there must be a corresponding data file in the `model/constraints/environment/` directory, except for `JFAC`, whose data file should go in the `model/constraints/photolysis/` directory. For more information, go to Sect. 4.2.1.

3.6.2 initialConcentrations.config

This file specifies the initial concentrations (in molecule cm^{-3}) of the chemical species. The file has two columns: the first column is the name of the initialized species, the second column is the corresponding concentration at the **model start time**, which is defined in the `model.parameters` file (Sect. 3.2). For example:

```
O3      1.213e+12
NO      378473308.14
NO2     86893908168.9
CH4     4.938e+13
```

Not all chemical species need to be initialized: those that are not listed in `initialConcentrations.config` are assumed to have an initial concentration of zero molecule cm^{-3}. It is also unnecessary to initialize the chemical species that are set to constant values or constrained to measured concentrations, i.e. those listed in `speciesConstant.config` and/or `speciesConstrained.config` (see below).

3.6.3 outputRates.config

This file lists the chemical species for which detailed production/loss rates (in molecule cm^{-3} s^{-1}) are required[6]. The file has one column, with one species per line. For example:

```
OH
HO2
NO3
```

The frequency of this output is controlled by **rates output step size** in the `model.parameters` file (Sect. 3.2). The corresponding output files – called `productionRates.output` and `lossRates.output` – are formatted to facilitate the rate of production and destruction analysis (ROPA/RODA) of the selected species. For more information, go to Sect. 4.5.1.

3.6.4 outputSpecies.config

This file[7] lists the chemical species for which the calculated concentration (in molecule cm^{-3}) is required. The file has one column, with one species per line. For example:

```
O3
NO2
CH4
OH
HO2
```

The frequency of this output is controlled by the **step size** parameter in the `model.parameters` file (Sect. 3.2). Constrained chemical species may be listed in `outputSpecies.config`, which can be useful for diagnostic and debugging purposes. Note that the photolysis rates, the environment variables, H_2O, and the RO2 sum are always output by the model (Sect. 4.5). Therefore, there is no `.config` file to set the output of these variables and they *should not* be listed in `outputSpecies.config`.

[6] In version 1.0 and earlier, these species were listed in two separate files: `productionRatesOutput.config` and `lossRatesOutput.config`.

[7] Called `concentrationOutput.config` in version 1.0 and earlier.

3.6.5 photolysisConstant.config

This file lists the photolysis rates that are set to constant values (Sect. 3.5.1). The file has three columns: the first column is the ID number of the photolysis rate, the second column is the value of the photolysis rate (in s^{-1}), the third column is the name of the photolysis rate. The ID numbers and the names of the photolysis rates follow the MCM designation. For example:

```
1     2.5e-5     J1
4     8.7e-3     J4
7     1.8e-3     J7
```

The photolysis rates that are not listed in `photolysisConstant.config` are automatically set to zero (Fig. 3.1). If no photolysis rate is set to constant, the file should be empty.

3.6.6 photolysisConstrained.config

This file[8] lists the photolysis rates that are constrained (Sect. 3.5.2). The file has one column, with one photolysis rate per line. The names of the photolysis rates follow the MCM designation. For example:

```
J1
J4
J7
```

If a photolysis rate is constrained, there must be a corresponding constraint file in the `model/constraints/photolysis/` directory (Sect. 4.2.1). The photolysis rates that are not listed in `photolysisConstrained.config` are calculated using the MCM parametrization (Fig. 3.1). If no photolysis rate is constrained, the file should be empty.

3.6.7 speciesConstant.config

This file[9] lists the chemical species that are set to constant concentrations. The file has two columns: the first column is the list of constant species, the second column is the corresponding concentration (in molecule cm^{-3}). For example:

```
CH4     4.4e+13
CO      2.5e+12
H2      1.2e+13
```

The chemical species that are set to a constant value do not need to be initialized: the values set in `speciesConstant.config` override those set in `initialConcentrations.config`. If no chemical species is set to constant, the file should be empty.

[8] Called `constrainedPhotoRates.config` in version 1.0 and earlier.
[9] Called `constrainedFixedSpecies.config` in version 1.0 and earlier.

3.6.8 speciesConstrained.config

This file[10] lists the chemical species that are constrained. The file has one column, with one species per line. If a chemical species is constrained, there must be a corresponding constraint file in the `model/constraints/species/` directory (Sect. 4.2.1). For example:

```
CH4
CO
H2
```

The chemical species that are constrained do not need to be initialized: the values set in `speciesConstrained.config` override those set in `initialConcentrations.config`. If no chemical species is constrained, the file should be empty.

[10] Called `constrainedSpecies.config` in version 1.0 and earlier.

Chapter 4

Model Execution

4.1 Box-Model

AtChem2 is a modelling tool to build and run chemical box-models for atmospheric chemistry and air quality studies [Sommariva et al., 2020]. An AtChem2 box-model requires two sets of user-supplied inputs: the chemical mechanism file and the configuration files, both are described in the following sections. Additionally, AtChem2 has the option to constrain a box-model to observations (see Sect. 4.2), in which case constraint data files mmust also be provided.

4.1.1 Mechanism file

AtChem2 is designed to use the Master Chemical Mechanism (MCM). Other chemical mechanisms can be used, as long as they are in the right format. By default, AtChem2 uses the FACSIMILE format, which is described in Sect. 3.1.1. The chemical mechanism must be provided as a text file with the extension `.fac`.

Since version 1.2.3, AtChem2 also accepts chemical mechanisms in KPP format: in this case, the `.kpp` file is automatically converted by the build scripts into a `.fac` file, which is then processed as any other FACSIMILE-formatted file. The mechanism file (`.fac` or `.kpp`) can be downloaded from the MCM website, as explained in Sect. 3.1.3, or it can be assembled manually. The user can modify the mechanism file with a text editor, if needed.

During the build process, the mechanism file is converted into a number of files in Fortran-compatible format and a pre-compiled shared library (`mechanism.so`). These files are created in the *`Shared Library Directory`* (by default, `model/sharedlib/`) and should not be modified by the user.

4.1.2 Configuration files

The configuration of the box-model is set via a number of text files, which can be modified with a text editor. Detailed information on the configuration files can be found in the corresponding sections:

- ◇ Model and solver parameters settings – go to Model Parameters and Solver Parameters.
- ◇ Environment variables settings – go to Environment Variables.
- ◇ Photolysis rates settings – go to Photolysis Rates.
- ◇ Model initialization, input and output – go to Config Files.

4.2 Constraints

An AtChem2 box-model can be set up in two modes:

Unconstrained : all variables are calculated by the model from the initial conditions, which are set in the configuration files.

Constrained : one or more variables are constrained, meaning that the solver forces them to given values at each timestep. The variables that are not constrained are calculated by the model from the initial conditions and from the constrained variables.

The constraint data must be provided as one text file for each constrained variable, with the format described below. The constraint data files are located in the *`Constraints Directory`* (by default, `model/constraints/`). The *`Constraints Directory`* contains three subdirectories, one for each type of constraint:

- ◇ `species/` for the chemical species.
- ◇ `environment/` for the environment variables.
- ◇ `photolysis/` for the photolysis rates.

All the default directories can be changed, as explained in Sect. 4.3 and 4.4 (see also Sect. 2.6.2), so that the user can create flexible and customized workflows.

4.2.1 Constrained variables

Environment variables

All environment variables, except `ROOF`, can be constrained. To do so, set the variable to `CONSTRAINED` in the file `environmentVariables.config` (Sect. 3.6.1) and create a file with the constraint data (Sect. 4.2.2). The name of the file must be the same as the name of the constrained variable without extension – e.g., `TEMP`.

Chemical species

Any chemical species in the chemical mechanism can be constrained. To do so, add the name of the species to `speciesConstrained.config` (Sect. 3.6.8) and create a file with the constraint data (Sect. 4.2.2). The name of the file must be the same as the name of the constrained chemical species without extension – e.g., `O3`.

Photolysis rates

Any photolysis rate in the chemical mechanism can be constrained. The photolysis rates are identified as `J<i>`, where `i` is the ID number assigned by the MCM to each photolysis reaction (Sect. 3.5). To constrain a photolysis rate, add its name to `photolysisConstrained.config` (Sect. 3.6.6) and create a file with the constraint data (Sect. 4.2.2). The name of the file must be the same as the name of the constrained photolysis rate without extension – e.g., `J4`.

Important Note: even though `JFAC` is an environment variable (Sect. 3.4), its constraint file must be located in the same directory as the photolysis rates constraint files.

4.2.2 Constraint files

All constraint data files are text files with two columns and no header: the first column is the time in seconds since midnight of the start date (in UTC, Sect. 3.2), while the second column is the value of the variable in the appropriate unit. For the chemical species the unit is molecule cm^{-3}; for the photolysis rates the unit is s^{-1}; for the units of environment variables, see Sect. 3.4. For example:

```
-1800   71.48
-900    73.21
0       74.393
900     72.973
1800    72.63
2700    72.73
```

```
3600    69.326
4500    65.822
5400    63.83
6300    64.852
```

The times in the first column of a constraint file do not have to be at regular intervals and can be negative. AtChem2 interprets negative times as *seconds before midnight of the start date* (Sect. 3.2). Data points with negative time can be useful to ensure the correct interpolation of the variables at the beginning of the model run. This is because the constrained data must cover at least the same amount of time as the intended model runtime, and preferably more, to avoid interpolation errors. For details, go to Sect. 4.2.3.

For example: if the model starts at 41400 seconds (day 1 at 11:30) and stops at 225900 seconds (day 3 at 14:45), then the first and the last data points of a constraint file must have a time of 41400 seconds (or lower) and 225900 seconds (or higher), respectively.

Important Note: the filenames of all constraint data files must match exactly the name of the constrained variable (including capitalization), without any extension.

4.2.3 Interpolation

Constraints can be provided at different timescales. Typically, the constraint data come from direct measurements and it is very common for different instruments to sample with different frequencies. For example, ozone (O_3) and nitrogen oxides (NO, NO_2) can be measured once every minute, but most hydrocarbons can be measured only once every 30–60 minutes.

The constraints can be used with their original timescales, or can be averaged so that they are all on the same timescale. Both approaches have advantages and disadvantages in terms of how much pre-processing work is required to prepare the data files, and in terms of model accuracy and integration speed: for a discussion of these issues, see Sommariva et al. [2020]. The choice of approach depends on the specific modelling objectives, as well as the available computing resources.

Whether all the constraints are on the same timescale or not, AtChem2 interpolates between data points using the interpolation methods selected in `model.parameters` (Sect. 3.2). The default interpolation method is piecewise linear, but piecewise constant interpolation is also available.

The photolysis rates and the environment variables are evaluated by the solver when needed – each is interpolated individually, and only when constrained. This happens every time the `mechanism_rates()` function is called from the `FCVFUN()` function, and is controlled by CVODE as it carries out the integration. In a similar way, the interpolation routine for the chemical

species is called once for each of the constrained species in `FCVFUN()`, plus once when setting the initial conditions of each of the constrained species.

The model start and stop times must fall within the time range of all constrained data to prevent interpolation errors or model crashes, as mentioned in Sect. 4.2.2 and illustrated below:

```
                  start            stop
model run           |----------------|
constraint 1    |---------------------------|
constraint 2       |------------------|
constraint 3     |-----------------------|
```

If constraint data are not supplied for the entire runtime interval, the final value of the constrained variable will be used for all times *before the first* data point and *after the last* data point. When this situation occurs, a warning is printed to the terminal for all data evaluations outside of the supplied time interval. For example:

```
error in piecewise linear interpolation
 4.3205895E+05 720  3.0000000E+04
error in piecewise linear interpolation
 4.3205895E+05 720  0.0000000E+00
```

Adjusting the **model start time** and/or the **number of steps** in the `model.parameters` file (Sect. 3.2) usually resolves this issue[1]. In any case, to avoid errors, it is good practice to always provide constraint data that include a short period before the start time and a short period after the stop time.

4.3 Build

The `build_atchem2.sh` script in the `build/` directory is used to process the chemical mechanism file (Sect. 4.1.1) and compile the model. The script generates a number of files describing the chemical mechanism in a Fortran-compatible format (`mechanism.*`) and the pre-compiled mechanism shared library (`mechanism.so`). These files are described in Sect. 3.1.5.

The `build_atchem2.sh` script must be run from the *`Main Directory`* and takes four arguments, which are provided using the following input flags[2]:

[1] This behaviour is not desired and likely to change in future versions to avoid the situation where the last value is used for all times before the first data point – see issue #294).

[2] Prior to version 1.3, the build script required three positional arguments – for details, see the documentation on the wiki.

1. Path to the chemical mechanism file (`.fac` or `.kpp`).
   flag: `--mechanism`
   There is no default for this argument, which must *always* be specified. It is suggested to keep the mechanism file either in the *Model Directory* or in the *Configuration Directory*.

2. Path to the *Configuration Directory*, for the model configuration files and the custom Fortran functions.
   flag: `--configuration`
   The default is `configuration/` in the *Model Directory*.

3. Path to the *Shared Library Directory*, for the Fortran mechanism files and the pre-compiled shared library.
   flag: `--shared_lib`
   The default is `sharedlib/` in the *Model Directory*.

4. Version of the MCM to be used.
   flag: `--mcm`
   The default is `v3.3.1`; other accepted values are `v3.2` and `v3.1`.

Both full and relative paths are acceptable, depending on the user's preference, and the order in which the flags are used does not matter. For example, a box-model with the default settings for the configuration, shared library and MCM version, using the example chemical mechanism file (`model/mechanism.fac`) can be built with the following command:

```
./build/build_atchem2.sh --mechanism=./model/mechanism.fac
```

As mentioned in Sect. 2.6.2, an installation of AtChem2 can have multiple *Model Directory*, each corresponding to a different model or project. This approach allows the user to work with multiple models at the same time using a single installation of AtChem2, and makes it easy to run batch simulations for sensitivity studies. The *Model Directory* can also be located outside the *Main Directory*, giving users great flexibility in the organization of their workflow.

Fig. 4.1 shows an example of a possible setup. A `Modelling/` directory within the user's home directory is designated for the modelling work: it contains an AtChem2 installation consisting of two directories: the *Main Directory* (`~/Modelling/AtChem2/`), which includes the default `model/` directory, and the *Dependencies Directory* (`~/Modelling/AtChem-lib/`). It also contains a project directory (`~/Modelling/Project_A/`) with two separate *Model Directory* (`model_1/` and `model_2/`), each with their own chemical mechanism, configuration, constraints and output. The project directory may correspond, for example, to a field campaign or to a set of related experiments. In this case, each model can be built independently from the *Main Directory* with the following commands:

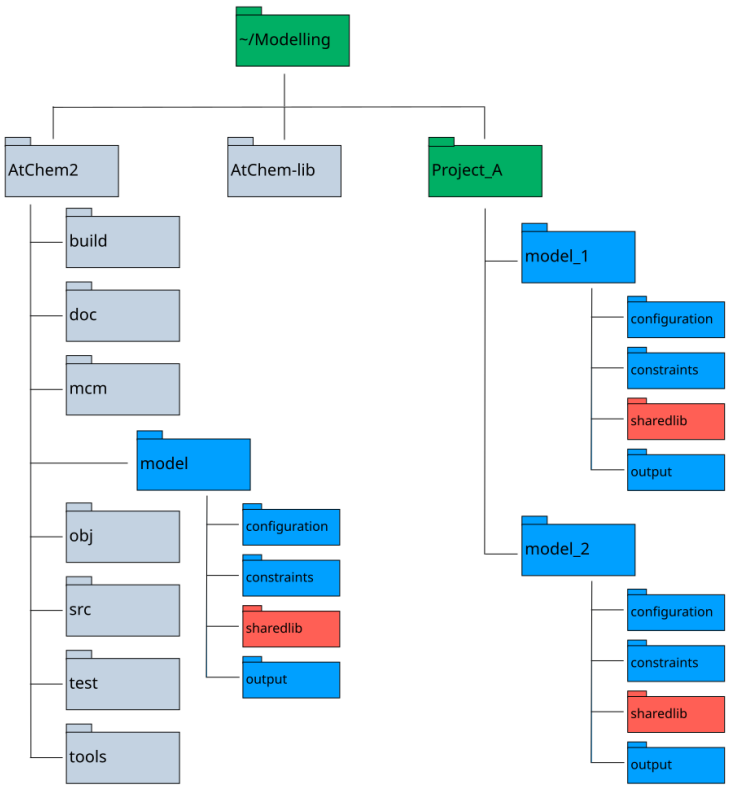


Figure 4.1: Example of a modelling setup, with one AtChem2 installation and one project directory containing two different box-models. Note that the user should not modify the content of the *Shared Library Directory* (`sharedir`).

```
./build/build_atchem2.sh --mechanism=../Project_A/model_1/mechanism.fac
                         --configuration=../Project_A/model_1/configuration/
                         --shared_lib=../Project_A/model_1/sharedlib/

./build/build_atchem2.sh --mechanism=../Project_A/model_2/mechanism.fac
                         --configuration=../Project_A/model_2/configuration/
                         --shared_lib=../Project_A/model_2/sharedlib/
                         --mcm=v3.2
```

Note that the second command in the example above changes the version of the MCM to v3.2: the `--mcm` flag affects only the choice of the RO2 sum and of the photolysis rates. It is the responsibility of the user to ensure that the chemistry in the `.fac` file corresponds to the correct version of the MCM.

There are many different ways in which the *Model Directory* can be organized, because each user has specific needs and personal preferences. For example, it is possible to keep all the constraint files related to a project

in one directory, thus avoiding the need to have identical `constraints/` directories in each *`Model Directory`*. As long as the correct paths are passed to the `build_atchem2.sh` script (and to the executable, see Sect. 4.4), the model will compile and run with the correct inputs.

The `build_atchem2.sh` script also handles model compilation, converting the Fortran code into the `atchem2` executable, which is dynamically linked to the shared chemical mechanism library (`mechanism.so`). If the Fortran code is not changed, compilation is required only once for a given mechanism file.

Changes to the configuration files do not require recompiling the model. Likewise, if constraint files are added, removed or modified, there is no need to recompile the model. This is because the model configuration and constraints are read by the executable at runtime. However, if the chemical mechanism file is changed, then the shared library (`mechanism.so`) needs to be recompiled.

If the user wants, or needs, to change the Fortran code (`src/*.f90`), then the executable needs to be recompiled. To recompile the model, the build script needs to be executed again[3]. The build script is able to detect whether the Fortran code, the chemical mechanism, or both have been modified and runs the appropriate commands. By compiling only when strictly necessary, this design reduces the frequency and duration of model compilations, facilitating batch model runs (e.g., for sensitivity studies).

4.4 Execute

The build process – described in Sects. 3.1.5 and 4.3 – creates an executable file called `atchem2` in the *`Main Directory`*. The executable takes up to eight arguments[4], corresponding to the *relative paths* (with respect to the *`Main Directory`*) of the model configuration, the chemical mechanism shared library, the constraint data files, and the model output. The arguments are passed to the executable via a series of input flags[5], each with a default value as described below:

1. Path to the *`Model Directory`*.
   flag: `--model`
   ↪ default: `model/`

2. Path to the *`Output Directory`*.

[3] If the recompilation fails, it is sometimes useful to run the command `make clean` from the *`Main Directory`* before rerunning the build script.

[4] The `--mcm` flag, which pointed to the directory with the MCM data files, was removed with version 1.3.

[5] Prior to version 1.2, the executable used a system of positional arguments instead of the input flags – for details, see the documentation on the wiki.

flag: `--output`
↪ default: `model/output/`

3. Path to the *Configuration Directory*.
   flag: `--configuration`
   ↪ default: `model/configuration/`

4. Path to the *Constraints Directory*.
   flag: `--constraints`
   ↪ default: `model/constraints/`

5. Path to the directory with the data files of constrained environment variables.
   flag: `--env_constraints`
   ↪ default: `model/constraints/environment/`

6. Path to the directory with the data files of constrained photolysis rates.
   flag: `--photo_constraints`
   ↪ default: `model/constraints/photolysis/`

7. Path to the directory with the data files of constrained chemical species.
   flag: `--spec_constraints`
   ↪ default: `model/constraints/species/`

8. Path to the *Shared Library Directory*.
   flag: `--shared_lib`
   ↪ default: `model/sharedlib/`

In addition, the input flag `--help` displays a help message explaining the usage of the command line arguments and of the input flags.

AtChem2 can be run simply by executing the command `./atchem2` from the *Main Directory*, in which case the executable will assume that all arguments have the default values. The equivalent command using the input flags explicitly is:

```
./atchem2 --model=model/
          --output=model/output/
          --configuration=model/configuration/
          --constraints=model/constraints/
          --spec_constraints=model/constraints/species/
          --env_constraints=model/constraints/environment/
          --photo_constraints=model/constraints/photolysis/
          --shared_lib=model/sharedlib/
```

Not all flags have to be used, and the order in which they are used does not matter. If a flag is omitted, the executable assumes the default value, based on a *hierarchical directory structure*. For example, the following command implies that a directory called `output/` is present inside the `model/`

directory, and that three directories called `species/`, `environment/` and `photolysis/` are present inside the `model/constraints/` directory:

```
./atchem2 --model=model/
          --configuration=model/configuration/
          --constraints=model/constraints/
          --shared_lib=model/sharedlib/
```

This approach gives the user considerable flexibility in the organization of the modelling work and makes it possible to run several models using the same executable with different configurations, constraints and chemical mechanisms. For example, the following commands run two models with the same chemical mechanism and two different configurations, and save the corresponding outputs in separate directories:

```
./atchem2 --configuration=model/configuration_1/
          --output=model/output_1/
          --shared_lib=model/sharedlib/

./atchem2 --configuration=model/configuration_2/
          --output=model/output_2/
          --shared_lib=model/sharedlib/
```

Only three input flags are specified in the above commands, and therefore the executable assumes the default values for the remaining input flags: in the example, this means that the *`Model Directory`* is `model/` and the *`Constraints Directory`* is `model/constraints/` for both model runs.

It is also possible to combine chemical mechanisms, configurations and constraints from different models. For example, using the setup shown in Fig. 4.1, the following command runs a model that has its own configuration but is constrained to the chemical species and environment variables of `model_1/` and to the photolysis rates of `model_2/`:

```
./atchem2 --configuration=model/configuration/
          --output=model/output/
          --shared_lib=model/sharedlib/
          --spec_constraints=../Project_A/model_1/constraints/species/
          --env_constraints=../Project_A/model_1/constraints/environment/
          --photo_constraints=../Project_A/model_2/constraints/photolysis/
```

While the model is running, diagnostic information is printed to the terminal. A successful model run completes with a message similar to the one shown in Sect. 2.4. The user has the option to redirect the terminal printout to a log file, using standard unix commands.

4.4.1 Execution on HPC

AtChem2 can be installed and run on high-performance computing (HPC) clusters, or supercomputers, which typically run some version of Linux/Unix. This is recommended, especially for models with long runtimes and/or many constraints.

Installation and setup on an HPC system are exactly the same as on a regular workstation, but each system has its own rules and guidelines. Therefore, it is not possible to give specific advice, and users should check the local documentation or ask the system administrator.

On most HPC systems, software execution is managed by a job scheduler, which requires a submission script. The script will allocate the requested resources[6], alert the user that the model run is finished, and save the terminal printout and any other message to log files. The syntax and format of the submission script depend on the type of job scheduler installed on the HPC system. A few examples of submission scripts can be found on the wiki.

4.5 Output

The model output is saved by default in the `model/output/` directory. The location can be modified by changing the arguments of the `atchem2` executable, as explained in Sect. 4.4 (see also Sect. 2.6.2). The frequency of the model output is determined by the following model parameters, which are set in the `model.parameters` file (Sect. 3.2):

- ◇ **step size** for the chemical species, environment variables, photolysis rates and diagnostic variables.
- ◇ **rates output step size** for the rate of production and destruction analysis of selected species (Sect. 4.5.1).
- ◇ **reaction rates output step size** for the reaction rates of all chemical reactions.

All AtChem2 output files are space-delimited text files with a one-line header containing the names of the variables. The first column of each file is the model time (`t`) in seconds since midnight of the **start date**[7]. The output files are:

- ◇ Environment variables and RO2 sum: `environmentVariables.output`

[6] AtChem2 does not support parallelization at present; this may change in future versions. See issue #536.

[7] Note that the **start date** is different from the **model start time**: the model start time indicates when the model begins its run and is in seconds since the midnight of the start date. For more information, go to Sect. 3.2.

- ◇ Concentrations of selected chemical species – see Sect. 3.6.4:
  `speciesConcentrations.output`
- ◇ Photolysis rates:
  `photolysisRates.output`
- ◇ Latitude, longitude, solar angles and related parameters:
  `photolysisRatesParameters.output`
- ◇ Loss and production rates (ROPA/RODA) of selected chemical species – see Sect. 3.6.3:
  `lossRates.output`, `productionRates.output`
- ◇ Concentrations of all chemical species, except the RO2 sum, at the end of the model run (**model stop time**):
  `finalModelState.output`
- ◇ Jacobian matrix (optional, see Sect. 3.2):
  `jacobian.output`
- ◇ Error messages and diagnostic variables:
  `errors.output`, `mainSolverParameters.output`

In addition to the `.output` files listed above, the reaction rates (in molecule cm^{-3} s^{-1}) of every reaction in the chemical mechanism are saved, by default, in the `output/reactionRates/` directory[8]: one file for each model step, with the name of the file corresponding to the output time in seconds. The files have no header, and the order of reactions is the same as in the chemical mechanism (under the "Reaction definitions" section – see Sect. 3.1.1).

4.5.1 ROPA/RODA

The reaction rates files in the `output/reactionRates/` directory are useful to analyze the model results and understand the chemical system, as well as for diagnostic purposes. The format, however, may be cumbersome to process. To simplify the analysis of the rates of production and destruction of specific chemical species (i.e. those listed in `outputRates.config`, see Sect. 3.6.3) AtChem2 also generates the two files `productionRates.output` and `lossRates.output`. These files contain the rates of the production and destruction reactions of the selected species in a human-readable format (in molecule cm^{-3} s^{-1}). The format is shown in Fig. 4.2.

The frequency of the ROPA/RODA output is controlled by the parameter **rates output step size**, as explained in Sect. 3.2. Note that the output files `productionRates.output` and `lossRates.output` can be very large

[8] Called `instantaneousRates/` prior to version 1.1.

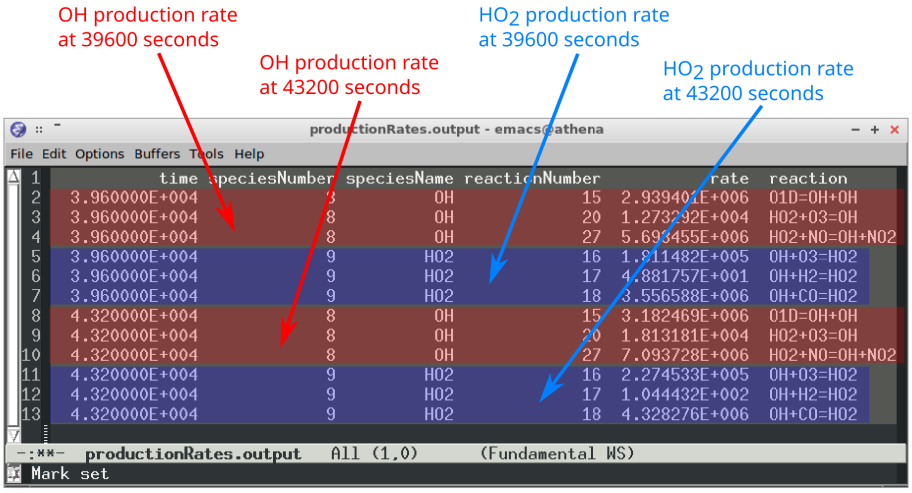


Figure 4.2: Format of the `productionRates.output` output file. The `lossRates.output` file has a similar format.

and complex, especially for species that are involved in many reactions and/or for long model runs. Therefore, it is recommended to limit the list of chemical species in `outputRates.config` only to those of special interest, and to keep the frequency of the output low (the default is 1 hour).

Important Note: although, in principle, a rate of production and destruction analysis for constrained chemical species can be done, the results should be interpreted with extreme caution. The user must be aware that the concentrations of these species are determined by the constraining data rather than by the chemical reactions.

Chapter 5

Model Development

5.1 General Information

AtChem2 is hosted on the GitHub developer platform. Two versions of the model are available in the repository at any given time:

Stable version : is indicated by a version number (e.g., `v1.0`) and can be downloaded from the Releases page. A **DOI** is assigned to each stable version via the Zenodo service.

Development version : is the `master branch` of the git repository and is indicated by a version number with the suffix "-dev" (e.g., `v1.1-dev`). It can be downloaded as an archive file or obtained with git – go to Sect. 2.2 for more information.

AtChem2 is under active development, which means that the `master branch` is sometimes a few commits ahead of the latest stable release. However, all code modifications are automatically run through the Testsuite before being merged into the `master branch`. Since the Testsuite is designed to ensure that changes to the code do not cause unintended behaviour or unexplained differences in the model results, even the development version of AtChem2 can be considered reliable for most use cases. Nevertheless, it is strongly suggested to use the **stable version** for "production runs" and publications, as it can easily be referenced via the associated DOI – see the discussion about traceability and reproducibility of computational models in Sommariva et al. [2020].

Feedback, bug reports, comments, suggestions, and feature requests are welcome. A list of open issues, known bugs and desired features can be found on the Issues page. Please check this page before reporting new issues/bugs; also note that the AtChem2 wiki lists some common known issues and contains suggestions on how to solve them.

Contributions and corrections to the user manual are also welcome. The manual is written in LaTeX, with figures in `.svg` format. The `.tex` files,

located in the `doc/latex/` directory, can be opened and modified using any text editor. To compile the `.pdf` file of the manual, use the script `doc/make_manual_pdf.sh`, then commit it together with the other changes[1]. Alternatively, the `.pdf` file is automatically compiled via GitHub Actions when a Pull Request (PR) that modifies one of the `.tex` or `.svg` files is opened.

The preferred way to contribute to the development of AtChem2 is to use the open-source git version control tool. Instructions on how to set up git, prepare and submit contributions to the AtChem2 repository can be found on the wiki. Contributors are strongly encouraged to follow the coding guidelines detailed in Sect. 5.3.

5.2 Testsuite

AtChem2 uses GitHub Actions for continuous integration testing. This programming approach ensures that changes to the code do not unintentionally modify the behaviour and the results of the software. The Testsuite is located in the `tests/` directory[2] and consists of a series of tests and short model runs that check the model functionality and calculations against known outputs. In addition, the Codecov service is used to ensure that the Testsuite covers a significant fraction of the codebase (aiming for >90% coverage) and a wide range of common configurations.

There are five types of tests, which can be executed from the *`Main Directory`* with the `make` command. Note that the *optional dependencies* must be installed in order to run the Testsuite – see Sects. 2.3.2 and 2.4.1 for details.

- The **Indent**, **Style** and **Python** tests check that the indentation and coding style of the Fortran and Python code are consistent with the guidelines:
  ↪ `make indenttest`
  ↪ `make styletest`
  ↪ `make pythontest`

- The **Unit** tests check that individual Fortran functions generate the expected outputs:
  ↪ `make unittests`

- The **Model** tests run a number of models with different configurations to check that they generate the expected outputs:

[1] The script requires a LaTeX installation and ImageMagick (v7.x).

[2] Called `travis/` in versions 1.0–1.2.2, when the TravisCI service was used to run the Testsuite.

↪ `make modeltests` and/or `make oldtests`[3]

- ◇ The command `make alltests` runs all the tests described above in succession.

The tests create `.log` files in the `tests/` directory and print a summary message to the terminal, while the Unit tests output the results directly to the terminal. If one or more tests fail, search the log files for detailed information.

The Testsuite is automatically run every time a PR is created or updated on the GitHub repository. The PR triggers a GitHub Actions workflow that runs the entire Testsuite on two architectures (Linux and macOS) using different versions of the GNU `gfortran` compiler. The workflow – described in the `.github/workflows/ci.yml` file – performs the following tasks on each architecture:

- ◇ Install `gfortran` and the AtChem2 dependencies using `apt-get` (Linux) or `Homebrew` (macOS).
- ◇ Build and run a model using the default chemical mechanism and configuration: PASS if it exits with status 0.
- ◇ Build and run the Indent, Style, Python tests: PASS if all tests pass.
- ◇ Build and run the Unit tests: PASS if all Unit tests pass.
- ◇ Build and run the Model tests: PASS if there are no differences from the reference output files (within tolerance).
- ◇ Build and run the Unit and Model tests, and upload the coverage reports to the Codecov service.

The Testsuite is successful only if all tests pass. This is indicated by the message "All checks have passed" on the GitHub PR page. Pull Requests should only be merged into the `master branch` once GitHub Actions has completed with passes on both architectures.

5.2.1 Adding new Unit tests

The Unit tests are in the `tests/unit_tests/` directory and require the FRUIT optional dependency (which needs Ruby v3.0 or earlier, Sect. 2.3.2). To add new Unit tests, follow the procedure outlined below:

[3] At present these tests are split into two groups, `modeltests` and `oldtests`, which will eventually be merged. Both groups of tests pass on Linux, but `modeltests` fail on macOS. See issue #522.

- ◇ The Unit test files are called `*_test.f90`. If the new Unit test to be added fits into an existing test file, edit that file – otherwise, create a new test file following the same naming pattern. It is suggested that Unit tests covering functions from the source file `xFunctions.f90` be named `x_test.f90`.
- ◇ The Unit test file must contain a module with the same name as the file – i.e. `x_test` – and it must include the statement "use fruit", plus any other required module.
- ◇ The module should contain a number of subroutines with the naming pattern `test_*`. These subroutines must take no arguments and, importantly, must not have any brackets for arguments – subroutine `test_calc` is correct, but subroutine `test_calc()` is wrong.
- ◇ Each subroutine should call one or more assert functions: usually, these are `assert_equals()`, `assert_not_equals()`, `assert_true()` and `assert_false()`. The assert functions act as the arbiters of pass or failure of the Unit test – each assert must pass for the subroutine to pass, and each subroutine must pass for the Unit tests to pass.
- ◇ The assert functions have the following syntax:

```
call assert_true( a == b , "Test that a and b are equal")
call assert_false( a == b , "Test that a and b are not equal")
call assert_equals( a, b , "Test that a and b are equal")
call assert_not_equals( a, b , "Test that a and b are not equal")
```

It is useful to make the last argument of the assert function a *unique* and *descriptive* message. When a Unit test fails, it is highlighted in the FRUIT output summary, and the message of the assert function is printed. Unique and descriptive messages thus enable faster and easier understanding of which test has failed, and perhaps why.

If these steps are followed, calling `make unittests` is enough to run all the Unit tests, including the new ones. To verify that the new tests have indeed been run and passed, check the output summary – there should be a line associated with each of the `test_*` subroutines in each test file.

5.2.2 Adding new Model tests

Each Model test (`$TESTNAME`) is contained in its own subdirectory inside the `tests/model_tests/` directory. A Model test requires the following files and directory structure:

```
|- $TESTNAME
|  |- configuration
|  |  |- customRateFuncs.f90           [*]
```

```
|  |  |- environmentVariables.config
|  |  |- initialConcentrations.config
|  |  |- mechanism.prod.cmp
|  |  |- mechanism.reac.cmp
|  |  |- mechanism.ro2.cmp
|  |  |- mechanism.species.cmp
|  |  |- model.parameters
|  |  |- outputRates.config
|  |  |- outputSpecies.config
|  |  |- photolysisConstant.config     [*]
|  |  |- photolysisConstrained.config  [*]
|  |  |- solver.parameters
|  |  |- speciesConstant.config        [*]
|  |  |- speciesConstrained.config     [*]
|  |- constraints      [**]
|     |- environment/  [**]
|     |- photolysis/   [**]
|     |- species/      [**]
|- output
|  |- reactionRates/
|  |- environmentVariables.output.cmp
|  |- errors.output.cmp
|  |- finalModelState.output.cmp
|  |- jacobian.output.cmp
|  |- lossRates.output.cmp
|  |- mainSolverParameters.output.cmp
|  |- photolysisRates.output.cmp
|  |- photolysisRatesParameters.output.cmp
|  |- productionRates.output.cmp
|  |- speciesConcentrations.output.cmp
|- $TESTNAME.{fac|kpp}
|- $TESTNAME.out.cmp
```

The files marked with `[*]` and the directories marked with `[**]` are optional, depending on the model configuration used in the test. If present, the directories marked with `[**]` should contain the relevant constraint files, according to the corresponding configuration files in `model/configuration/` (see Sect. 4.2 for details). The file `$TESTNAME.out.cmp` should contain an exact copy of the expected terminal printout.

Each Model test is briefly described in the `tests/model_tests/INFO.md` file, which should be updated after a new test is added. New tests added to the `tests/model_tests/` directory are automatically picked up when the commands `make modeltests` or `make alltests` are run from the *Main Directory*.

5.3 Style Guide

In order to make the AtChem2 code more readable and easier to maintain, the Fortran source code should follow a consistent style (Sect. 5.3.1). Two Python scripts are available to check and, if necessary, correct the Fortran code:

- ◇ `fix_style.py` edits a Fortran file in place to make the code consistent with the style recommendations.
- ◇ `fix_indent.py` works in a similar way, but looks only at the indentation level of each line of code.

These scripts are in the `tools/` directory and are executed from the *`Main Directory`* with the following commands:

```
python tools/fix_style.py src/filename.f90
python tools/fix_indent.py src/filename.f90
```

It is important to keep in mind that these scripts are *not infallible* and, therefore, it is strongly recommended to always have a backup of the original Fortran file to revert to, in case of wrong edits. This can be done by passing two arguments to the script instead of one: the second argument sends the script output to another file, leaving the original file untouched.

Both scripts are also used in the Testsuite to run the Style and Indent tests (Sect. 5.2): each script is run over each source file and the output is sent to a `.cmp` file. If all `.cmp` files match the original files, the test passes.

5.3.1 Style recommendations

The following guidelines apply to the Fortran code only. For Python, AtChem2 adopts the recommendations of the black formatter (v26). Adherence to the Python code guidelines can be checked by executing the command `make pythontest` from the *`Main Directory`* (see Sect. 5.2 for details). There are no style guidelines for the AtChem2 shell scripts; the only requirement is that the code has to be sh-compliant, which can be verified with the shellcheck utility – installed locally or online at https://shellcheck.net.

General principles

- ◇ All code should be organized in a **module structure**, except for the main program. There is one further exception: due to an issue with linking to CVODE, the functions `FCVFUN()` and `FCVJTIMES()` are placed within the main file (`atchem.f90`).

- ◇ All code should be written in free-form Fortran, compliant with the Fortran 90/95 standard; the source files should have the extension `.f90`.
- ◇ Always use two spaces to indent blocks.
- ◇ At the top of each file there should be a header indicating the author(s), date, and purpose of the code. If applicable, acknowledgements to other contributors should be added.
- ◇ Always comment a procedure with a high-level explanation of what that procedure does.
- ◇ There are no specific guidelines for comments, although common sense applies, and any code within the comments should broadly follow the rules below.

Specific recommendations

- ◇ All **keywords** should be lowercase – for example: `if then`, `call`, `module`, `integer`, `real`, `only`, `intrinsic`. This includes the `(kind=XX)` and `(len=XX)` statements.
- ◇ All **intrinsic** function names should be lowercase: e.g., `trim`, `adjustl`, `adjustr`.
- ◇ The **relational operators** should use `>=` and `==` rather than `.GE.` and `.EQ.` – they should be surrounded by a single space.
- ◇ The `=` operator should be surrounded by one space when used as assignment, except in the cases of the `(kind=XX)` and `(len=XX)` statements, where no spaces should be used.
- ◇ The **mathematical operators** (`*`, `-`, `+`, `**`) should be surrounded by one space.
- ◇ Numbers in scientific notation should have no spaces around the `+` or `-` signs: e.g., `1.5e-9`.
- ◇ The names of the **variables** should begin with lowercase, while those of the **procedures** (subroutines and functions) should begin with uppercase. An exception to this rule are **third-party functions**, which should be uppercase. Both CamelCase and underscores are acceptable for multiple-word identifiers, but be consistent.
- ◇ All **modules** should include the `implicit none` statement.
- ◇ All **variable declarations** should include the `::` notation.

- ◇ The **dummy arguments** of a procedure should include an `intent` statement in their declaration.
- ◇ The following rules apply to **brackets**:
  - ○ Opening brackets should not have a space before them, except for `read`, `write`, `open` and `close` statements.
  - ○ All `call` statements and the definitions of all procedures should contain one space before the first argument and one after the last argument inside the brackets:

    ```
    call function_name( arg1, arg2 )
    subroutine subroutine_name( arg1 )
    ```

  - ○ Function calls and array indices should not have spaces before the first argument or after the last argument inside the brackets.

Chapter 6

Credits, Acknowledgements & Funding

6.1 Credits

AtChem2 has been developed at the University of Leicester by:

- Sam Cox
- Roberto Sommariva (also at the University of Birmingham)

Additional code and/or documentation have been contributed by (in alphabetical order):

- James Allsopp
- Will Drysdale
- Maarten Fabré
- Alfred Mayhew
- Killian Murphy
- Beth Nelson
- Mike Newland
- Marios Panagi

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- Chris Martin

- Kasia Borońska
- Jenny Young
- Peter Jimack
- Mike Pilling

Evaluation and testing of AtChem-online were conducted by Andrew Rickard (NCAS/University of York) and Monica Vázquez-Moreno (CEAM); technical support was provided by David Waller (University of Leeds).

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- National Centre for Atmospheric Science (NCAS)

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◇ University of Birmingham, Research Software Group

◇ University of Leicester, ReSET programme

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